

SUCCESSOR Ω — ASTERION Ω

The Scientific Industrial Estate

Executable world models. Decision-relevant physics.
Strategic verification. Resource-bounded expansion.



CONCEPTUAL HORIZON / AI-GENERATED ART. NOT AN ENGINEERED DESIGN.

Publication record and reading contract

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This monograph turns an executed customer simulation into an inspectable scientific and institutional argument. It preserves the experiment's negative competitive result. It adds explicit mathematical conditions for useful learning, physically consistent decisions, credible verification incentives and resource-bounded expansion. The added propositions are derived here as specialized consequences of established methods, not claimed as previously unknown mathematics.

The evidence boundary

Executed: local neural training, symbolic search and revision, thermal-design decisions and numerical checks. **Derived:** conditional propositions and new finite mathematical calculations. **Assumed:** fictional ownership, programme scale, prices and the separate 2.5% future process improvement. **Prospective:** real materials, independently qualified Specialist ASI, live deployment and stellar infrastructure. These classes must not be merged.

The architectural source explicitly separates capability, proof and authority, and defines Specialist ASI as a mission-bounded admission status. Its diagrams on pages 12 and 21 respectively organize the institutional planes and separate a frozen serving release from a learning successor. This paper adopts those distinctions; the ASTERION customer extension does not execute the final admission process.[1]

Owner route. Read the decision brief, Sections 1, 7, 13–15 and the qualification matrix. **Research route.** Read Sections 3–6, 8–11 and the proof appendices. **Implementation route.** Read the lineage table, exact protocol, source map and reproduction commands.

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Abstract

SUCCESSOR Ω —ASTERION Ω studies a customer-controlled institution that converts observations into executable hypotheses, hypotheses into physical design choices, and qualified outcomes into the means of financing further research. A retained experiment uses the unmodified scientific engine of NEURAL-SYMBOLIC SUCCESSOR Ω v10.2 and a separately authored thermal/economic adapter. Eight cohorts train eight local neural guides and select eight symbolic emissivity models after 10,696 native candidate fits. Neural ablation changes the proposed observation in every cohort. A common, paid-remediation planner compares 2,048 final decisions against a conservative incumbent and a development-selected Beta.

At an assumed 100-GW heat-rejection programme scale, the modeled cost contrast against the incumbent is CA\$ 6.692 billion. The more relevant mean edge over Beta is only CA\$ 1.194 million before incremental overhead and CA\$ -0.806 million after it; an eight-cohort bootstrap interval is [CA\$ -13.325, CA\$ 12.085] million. A separate degradation shift creates 599 initial capacity shortfalls. Thus nominal prediction success and accounting consistency do not establish competitive Alpha or adverse-regime qualification. These losing and failure results are part of the contribution.[2]

The formal analysis connects spectrally weighted thermal radiation, radiosity, exergy dissipation and robust thermal barriers to decision-relevant uncertainty. It derives conservative area and regret bounds, a simultaneous comparator certificate, inspection-game deterrence with false-positive and enforcement effects, proper-score elicitation, correlated scheduling, and the exact potential-game/Gibbs relation under finite asynchronous logit dynamics. A delayed capital-to-capacity model makes positive unsaturated growth conditional on reproduction exceeding retirement while preserving cash, material, energy and throughput constraints. These models are not asserted to be implemented production controls.

A separate underwriting case asks what a *new*, independently established 2.5% process improvement over refreshed Beta could support: CA\$ 197.827 million of conditional first-programme contribution after CA\$ 283 million in stated costs and reserves. No such improvement was discovered here. The long horizon is a scientific industrial estate extending its feasible missions toward stellar-scale productive networks, with power, heat, area, mass and signal delay explicit. No unrestricted discovery, independent Specialist ASI, real customer Alpha or production authority is claimed.

Keywords. Neural-symbolic discovery; emissivity; robust design; inspection games; correlated equilibrium; strategic learning; mission Alpha; institutional succession; productive capacity.

Decision brief: what is worth owning?

The owner’s objective

Own the ability to discover what deserves to become a productive asset—and retain enough evidence to replace the institution’s own understanding when reality disagrees.

The most valuable use of the present release is a disciplined research-and-design programme. The customer buys neither a guarantee of superiority nor a permission to automate consequential action. The reusable asset is a transparent scientific workflow, an archive of fitted and rejected programs, and the practical ability to reproduce a decision comparison. Its value may be learning, reduced waste or improved design; each financial claim still needs its own counterfactual.

Decision-relevant quantity	Retained result
Competitive mean after incremental overhead	CA\$ -0.806 million
First-programme mean after research allowance	CA\$ -3.806 million
Positive cohorts after overhead	4 / 8
Initial shortfalls under shifted degradation	599 / 2 048
Current customer Alpha credited	CA\$ 0
New improvement needed to break even in the separate scenario	1.4714%

Keep: the trained artifacts, portable comparator, negative evidence, accounting checks and next-experiment rationale. **Do not promote:** the current proprietary Alpha claim or shifted-regime deployment. **Investigate next:** a real material/process improvement that expands the available design set and survives a refreshed equal-information comparison.

The CA\$ 6.692 billion incumbent contrast, the CA\$ -0.806 million competitive result and the CA\$ 197.827 million conditional future case answer different questions. They are neither additive nor interchangeable. A successful institution can reject its own current candidate without abandoning the larger mission.

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1 The institution that learns what to build

OWNER SCENARIO / FICTIONAL CONTEXT, EXECUTED RESEARCH CORE

Dr. Adrienne Valcourt, a fictional physicist-engineer, controls an industrial group with CA\$ 180 billion in productive assets. Energy, computation, precision manufacturing and aerospace share a recurring problem: capital is available before the group knows which physical design deserves it. Her mandate is not to maximize model size. It is to acquire understanding that changes a consequential decision, establish what that change is worth, and retain the method across generations of suppliers and researchers.

The first constituted mission

Choose a material recipe and effective emitting area that meet a stipulated thermal requirement at defensible programme cost. Expose the observations that could defeat the model. Charge for shortfalls and remediation. Compare against a capable alternative with the same information and feasible options. Enable no apparatus, procurement or production actuation.

The thermal programme is authored at 100 GW of aggregate *heat rejection*, not electrical generation. Its scale magnifies economic consequences for inspection; it does not establish a built installation, useful lifetime, market price or contracted customer. The deployment boundary remains research-only.

1.1 An owned process, not a permanent champion

The enduring objects are the Mission Constitution, evidence contracts, model programs, trained parameters, comparator configurations, tests, decision ledgers and revision history. The present simulation produces many of those objects. It does not establish legal ownership by merely exporting files, nor does it activate the complete institutional control plane. The original licence and rights boundaries remain applicable.[1, 2]

The optimal owner buys the strongest suitable Beta whenever ownership cannot earn its full cost. She sponsors new research when a genuine change to the physical or process opportunity set could create an advantage that others cannot immediately reproduce. A discovery institution is economically rational only while its expected information and decision benefits justify its experiment, integration and governance burden.

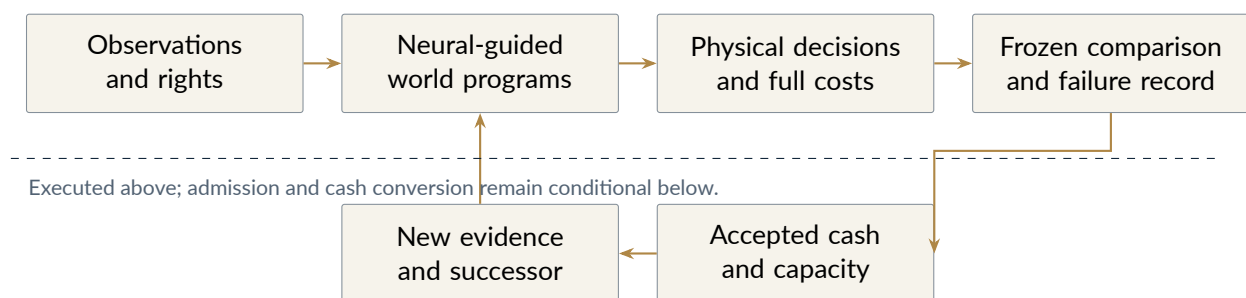


Figure 1. The discovery-to-capacity route. The lower branch is a prospective institutional continuation, not an executed financial or production connector.

2 What the retained experiment actually executes

RETAINED EMPIRICAL METHOD / SAME-AUTHOR SYNTHETIC EVIDENCE

2.1 Lineage and division of implementation

Layer	Actual role
SUCCESSOR Ω v10.2	Unmodified native scientific search, coefficient fitting, bounded grammar expansion, neural training, proposal scoring and model identity.
ASTERION 1.0	Customer-side thermal mission, synthetic data, mission-utility reselection, finite domain screen, shared residual margin, recipe/area planner and economic checker.
This monograph	Retained-record reanalysis, formal physics/game/economic conditions and finite numerical checks. No modified production runtime.

The study preserves 55 first-party source payload files without modification. An upstream record reports 840 runtime tests and 106 research tests; those counts are not relabeled as fresh ASTERION qualification. The present publication reruns the 24 customer-extension tests and the eight-cohort experiment in the available authoring environment, separately from the new mathematical calculations.[2, 3, 4]

2.2 The authored material and observations

The reference is a scalar stationary response, not a discovered material law or a full dynamical world model:

$$\varepsilon_{\star}(u, d) = 0.62 + 0.28u - 0.16u^2 - 0.09d - 0.055ud, \quad (u, d) \in [0, 1] \times [0, 2]. \quad (2.1)$$

Recipe u , exposure d and emissivity are dimensionless. Independent coupon noise has standard deviation 0.002. The designer chose a polynomial-compatible world within the engine’s expressive reach. Predictors do not receive oracle coefficients at inference, but designer knowledge still limits independence and generalization.

Partition per cohort	Rows	Use
Training	73	24 initial, 48 additional, one active noiseless probe
Selection	48	16 initial and 32 additional
Calibration	64	Shared finite residual-margin rule
Final holdout	128	Nominal prediction evaluation
Native “stress” field	64	Additional nominal observations, not shifted-regime evidence
Development policy cases	96	Customer-side utility selection
Final policy cases	256	Paired comparison and separate shift challenge

Eight cohorts use seed $881034 + 1000k$, $k = 0, \dots, 7$. The retained protocol fixes the candidate grammar, selectors and economic scale before final case generation. It was not externally preregistered. An exploratory pilot, a pre-fit schema abort and subsequent metadata/checker corrections are disclosed in the development log. Replaying the same cohorts is repeatability, not new independent evidence.

2.3 Search, screening and serious alternatives

Native search visits expression subsets with a maximum degree of three and six terms. The customer adapter refits visited structures on training data, screens 121 endpoint-inclusive domain points, calibrates a residual

margin and selects by mean development decision cost. Original native defaults and customer-selected artifacts remain separate. Screening finite points is not a global range proof.

Beta receives the same observations, margin rule, 21-recipe menu and development objective. Thirteen alternatives comprise polynomial ridge degrees one through four, each with regularization 10^{-4} , 10^{-2} , 1, plus an ExtraTrees model. All selected Betas are portable polynomial models. This is a serious finite comparison, not exhaustive access to every specialist or current commercial service.

3 A hypothesis that changes a decision

EXECUTED LEARNING / NOT NOVEL PHYSICAL LAW DISCOVERY

Cohort one initially selects

$$\hat{\varepsilon}_0(u, d) = 0.15136092u + 0.63657398 \cos d. \quad (3.1)$$

A trained neural surrogate participates in scoring a candidate measurement. Its selected probe is $(u, d) = (0.3090381, 1.8482457)$, where the parent error is approximately 0.62108. Removing the neural signal changes the selected probe in all eight trials. That is an ablation of computational participation, not proof of optimal experimental design.

The customer-selected successor in the first cohort is

$$\begin{aligned} \hat{\varepsilon}_1(u, d) = & 0.35456635 - 0.088555234d - 0.05746792du \\ & + 0.25975967 \sin u + 0.25411387 \cos u + 0.01090814e^u. \end{aligned} \quad (3.2)$$

The program is directly used in the recipe-and-area computation. Its fitted transcendental terms are an approximation in the permitted domain, not a causal explanation of a coating. A useful symbolic representation can remain scientifically provisional. Figure 2 compares the frozen programs with the authored reference on one declared slice.

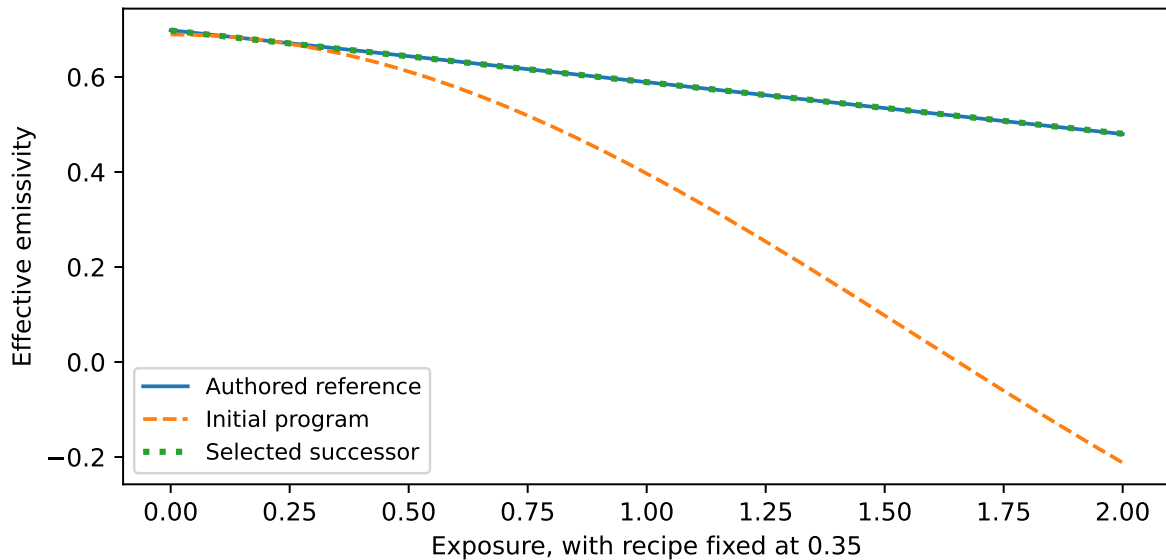


Figure 2. Retained programs at fixed recipe $u = 0.35$. Curves are recomputed from frozen artifacts; the reference is synthetic.

Across cohorts there are 10,696 native candidate fits, 104 Beta fits and eight trained neural guides. Each guide has 81 fitted weights and biases plus normalization state. The result demonstrates bounded neural-symbolic computation without calling an external neural provider. It does not demonstrate unrestricted theory discovery, autonomous sensor invention or a laboratory connector.[2]

Symbolic regression, sparse identification and program-synthesized world models provide relevant precedent. Their methods are distinct: a static fitted emissivity map is not a dynamical SINDy model, and the local numerical engine is not WorldCoder. ASTERION uses a common discipline—explicit programs whose consequences can be tested—without attributing another system’s benchmark gains to this product.[5, 6, 7]

4 Thermal physics: from one scalar to a physical envelope

NEW FORMAL TREATMENT / NOT A CLAIM OF IMPLEMENTED PLANT CONTROL

4.1 Spectral weighting matters

The emitted hemispherical blackbody spectral exitance is

$$M_\lambda(T) = \frac{2\pi hc^2}{\lambda^5} \left[\exp\left(\frac{hc}{\lambda k_B T}\right) - 1 \right]^{-1}. \quad (4.1)$$

A temperature-dependent effective emissivity is a weighted integral,

$$\varepsilon_{\text{eff}}(T) = \frac{\int_0^\infty \varepsilon_\lambda(T) M_\lambda(T) d\lambda}{\sigma T^4}. \quad (4.2)$$

Solar absorptance and infrared emittance generally use different spectral weights. Even where reciprocity and thermal-equilibrium assumptions justify equality of absorptance and emissivity at a given wavelength and direction, they do not make broadband solar absorptance equal to broadband thermal emittance. NASA’s thermal-control reference emphasizes those separate roles.[8]

ASTERION fits only an effective scalar at a stipulated temperature. A transferable material institution would need spectral, angular, temperature, exposure and process dependence, with instrument calibration and identifiable observations. Those measurements are proposed requirements, not hidden data in the delivered study.

4.2 Radiosity and the missing geometry

For a diffuse gray enclosure, surface radiosity J_i and irradiation G_i satisfy

$$J_i = \varepsilon_i \sigma T_i^4 + (1 - \varepsilon_i) G_i, \quad G_i = \sum_j F_{ij} J_j, \quad Q_i = A_i (J_i - G_i). \quad (4.3)$$

View factors must respect $F_{ij} \geq 0$, enclosure closure $\sum_j F_{ij} = 1$ and reciprocity $A_i F_{ij} = A_j F_{ji}$. For a closed two-surface diffuse-gray exchange model, the resistance form is

$$Q_{12} = \frac{\sigma(T_1^4 - T_2^4)}{\frac{1-\varepsilon_1}{A_1 \varepsilon_1} + \frac{1}{A_1 F_{12}} + \frac{1-\varepsilon_2}{A_2 \varepsilon_2}}. \quad (4.4)$$

This is a stated diffuse-gray, isothermal-surface model, not a general optical transfer solution. ASTERION’s original relation is its much simpler ideal sink limit:

$$Q = \varepsilon \sigma A (T^4 - T_s^4), \quad F = 1. \quad (4.5)$$

The numerical constant is $\sigma = 5.670374419 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$, rounded.[9] Effective emitting area is not necessarily deployed footprint. Two-sided emission, mutual irradiation and incoming sunlight must be resolved before an area is interpreted as hardware.

4.3 Transient energy and thermodynamic admissibility

A proposed thermal-network extension would carry state $T_i > 0$, heat capacity $C_i > 0$, symmetric conductance $K_{ij} \geq 0$ and net external power p_i :

$$C_i \dot{T}_i = p_i + \sum_j K_{ij}(T_j - T_i) + \sum_j q_{ji}^{\text{rad}}. \quad (4.6)$$

Antisymmetric exchange $q_{ji}^{\text{rad}} = -q_{ij}^{\text{rad}}$ preserves energy internally. For an unforced conductive network, define availability relative to a fixed $T_0 > 0$:

$$\mathcal{V}(T) = \sum_i C_i [T_i - T_0 - T_0 \log(T_i/T_0)] \geq 0. \quad (4.7)$$

Proposition 4.1 (Dissipation of thermal availability). For constant heat capacities, symmetric nonnegative conductances and no input or radiation term,

$$\dot{\mathcal{V}} = -T_0 \sum_{i < j} K_{ij} \frac{(T_i - T_j)^2}{T_i T_j} \leq 0. \quad (4.8)$$

Appendix C supplies the derivation. The same sign follows for passive pairwise radiation with the heat flow directed from hotter to colder. The publication checks the identity on 2,000 random finite conductive networks; that numerical check is not a proof of a manufactured system. A symbolic hypothesis violating passive heat-flow direction should fail before its apparent economic benefit reaches the planner.

4.4 Radiant power is not useful work

For an ideal blackbody radiation field, the energy and entropy fluxes scale as σT^4 and $\frac{4}{3}\sigma T^3$. Relative to a blackbody environment T_0 , their ideal exergy combination is

$$\dot{X} = \sigma T^4 \left[1 - \frac{4}{3} \frac{T_0}{T} + \frac{1}{3} \left(\frac{T_0}{T} \right)^4 \right]. \quad (4.9)$$

This is a blackbody reference identity, not the efficiency of a solar collector, arbitrary diluted spectrum or gray film. Spectral and directional radiation entropy requires additional treatment.[10] It prevents the paper from silently identifying captured radiation, electricity, low-temperature heat and cash as one interchangeable quantity.

5 Decision-relevant uncertainty and paid physical consequences

5.1 The executed decision rule

For each recipe the adapter constructs $\underline{\varepsilon} = \hat{\varepsilon} - m$ with $m = \max(0.004, \max |\text{calibration residual}|)$. It admits finite predictions with $0.1 < \underline{\varepsilon} < 1$, chooses the cheapest of 21 recipes and sets normalized area $a = 1/\underline{\varepsilon}$. Its authored area price is

$$p(u) = 240 + 45u + 35u^2 \quad [\text{CAD}/\text{m}^2]. \quad (5.1)$$

After the synthetic reference supplies actual ε , the ledger computes

$$\begin{aligned} z &= a\varepsilon, & s &= (1 - z)_+, & a_{\text{extra}} &= s/\varepsilon, \\ C &= ap(u) + 1.6a_{\text{extra}}p(u), & z_{\text{final}} &= (a + a_{\text{extra}})\varepsilon. \end{aligned} \quad (5.2)$$

This preserves a requested–delivered–paid distinction. A deficit remains an adverse initial event even if the toy ledger can purchase enough area to repair its arithmetic. It does not demonstrate real-time repair or admissibility in a safety-critical mission.

Here C is normalized cost per square metre of reference area; multiplying by A_{ref} gives programme CAD. Final exposure cases are alternative scenarios for that same programme, not additional 100-GW installations to be summed.

5.2 A conservative certificate under a stronger assumption

Proposition 5.1 (Area feasibility and excess area). Assume the actual emissivity obeys $\varepsilon \geq e_{\min} > 2\delta$ and the predictor satisfies a *uniform* error bound $|\hat{\varepsilon} - \varepsilon| \leq \delta$ over all relevant choices. Set $\underline{\varepsilon} = \hat{\varepsilon} - \delta$, $s_T = \sigma(T^4 - T_s^4) > 0$ and $A_R = Q/(s_T \underline{\varepsilon})$. Then

$$\varepsilon s_T A_R \geq Q, \quad 1 \leq \frac{A_R}{A_\star} \leq \frac{e_{\min}}{e_{\min} - 2\delta}, \quad A_\star = \frac{Q}{s_T \varepsilon}. \quad (5.3)$$

The guarantee concerns a declared model class and domain. The retained maximum-calibration-residual rule does not establish its uniform premise, especially after candidate reselection using the same calibration set. The paper’s 1,200 finite arithmetic checks illustrate the conditional inequality; they do not convert finite calibration into a global certificate.

Proposition 5.2 (Prediction-to-design regret). On a finite common design set, suppose both true and estimated emissivities are at least $e_0 > 0$, unit prices are at most $p_{\max}$, and $|\hat{\varepsilon} - \varepsilon| \leq \delta$. For cost $C(u) = p(u)Q/(s_T \varepsilon(u))$ and an exact minimizer $\hat{u}$ of the corresponding estimated cost,

$$C(\hat{u}) - \min_u C(u) \leq \frac{2p_{\max} Q \delta}{s_T e_0^2}. \quad (5.4)$$

This is a clean sensitivity result, not a bound for every feature of the executed paid-remediation objective. Physical constraints, forecast-dependent feasible sets, procurement discretization and remedy timing may require additional terms. Its practical lesson is that accuracy is valuable in proportion to the sensitivity of the decision, not just the average prediction score.

5.3 Why marginal coverage is not a selected-design guarantee

Split conformal prediction can supply finite-sample marginal coverage under exchangeability with an appropriately fixed predictor and calibration protocol.[11] A material recipe selected *because* its lower estimate looks cheapest is not automatically an exchangeable random test point. Uniform bands, simultaneous finite-menu correction, policy-level calibration, or protected prospective decision trials are different possible remedies. None is silently asserted for ASTERION 1.0.

6 Scientific discovery as a decision problem

6.1 What neural disagreement actually does

The native experiment-proposal heuristic uses a finite admissible pool. For symbolic predictions $f_j(x)$, neural prediction $n(x)$, cost $c(x) > 0$ and configured neural weight w , its score is

$$S(x) = \frac{\text{Var}_j[f_j(x)] + w(n(x) - \bar{f}(x))^2}{c(x)}. \quad (6.1)$$

The implementation's tie-break favors lower declared cost, smaller normalized excursion from the domain center, then identifier. The variance is empirical model disagreement, not a calibrated posterior. With one symbolic model it is zero, leaving neural-symbolic disagreement to distinguish probes. This is useful exploration; it is not a measured information gain or a theorem of scientific optimality.

6.2 A proposed value-of-information target

For future work, let θ represent uncertain material/process behavior, D the acquired evidence, a a design decision and $U(a, \theta)$ its full utility. The value of a proposed experiment x with observation Y_x is

$$\text{EVSI}(x) = \mathbb{E}_{Y_x|D} \left[\max_{a \in \mathcal{A}} \mathbb{E}[U(a, \theta) \mid D, Y_x] \right] - \max_{a \in \mathcal{A}} \mathbb{E}[U(a, \theta) \mid D] - c(x). \quad (6.2)$$

The before/after action set must be held fixed or its change explicitly priced. A safe no-experiment option yields zero net information value. The criterion can reject an intellectually interesting measurement when it cannot alter the decision enough to justify cost, delay or exposure. Information-directed sampling offers a related exploration-exploitation perspective based on decision regret and mutual information; it is not the implemented heuristic of Equation (6.1).[12]

For a differentiable observation model $y = f_\theta(x) + \eta$, $\eta \sim \mathcal{N}(0, \sigma_\eta^2)$, a local Fisher contribution is $I_x = \sigma_\eta^{-2}(\partial_\theta f)(\partial_\theta f)^\top$. A log-determinant information criterion may prefer identifiable experiments, but it depends on local parameterization and model adequacy. An unmodeled degradation mechanism can defeat both a narrow Fisher design and an overconfident posterior.

6.3 A route to broader discovery without relabeling scope

The scientific frontier is a sequence of executable extensions: new typed observables; temporal and causal models; conservation-aware operators; controlled apparatus adapters; richer hypothesis families; and adversarial tests whose outcomes may invalidate the whole representation. Each addition needs an observation contract, dimensional rules, resource limits and a falsification path. A claim of unrestricted discovery would exceed the current finite grammar and supplied observables. The research programme can pursue broader discovery while recording exactly what each release has earned.

7 Results: the incumbent contrast is not competitive Alpha

RETAINED RESULTS / LOSING AND ADVERSE OUTCOMES PRESERVED

The economic programme scales normalized design costs by

$$A_{\text{ref}} = \frac{10^{11}}{\sigma(450^4 - 3^4)} = 43\,006\,906.935 \text{ m}^2. \quad (7.1)$$

This is an authored 100-GW design programme, not annual revenue or installed electrical capacity. Mean costs are CA\$ 25.924 billion for the conservative incumbent, CA\$ 19.233 billion for selected Beta and CA\$ 19.232 billion for ASTERION before its extra overhead. Beta captures nearly the entire improvement over the incumbent.

After an assumed CA\$ 2 million incremental overhead, the competitive mean is CA\$ -0.806 million. Including a further CA\$ 3 million research allowance makes the first-programme mean CA\$ -3.806 million. No such monetary amount is a measured invoice. Real integration, financing, tax, lifetime and mission costs could change it materially.

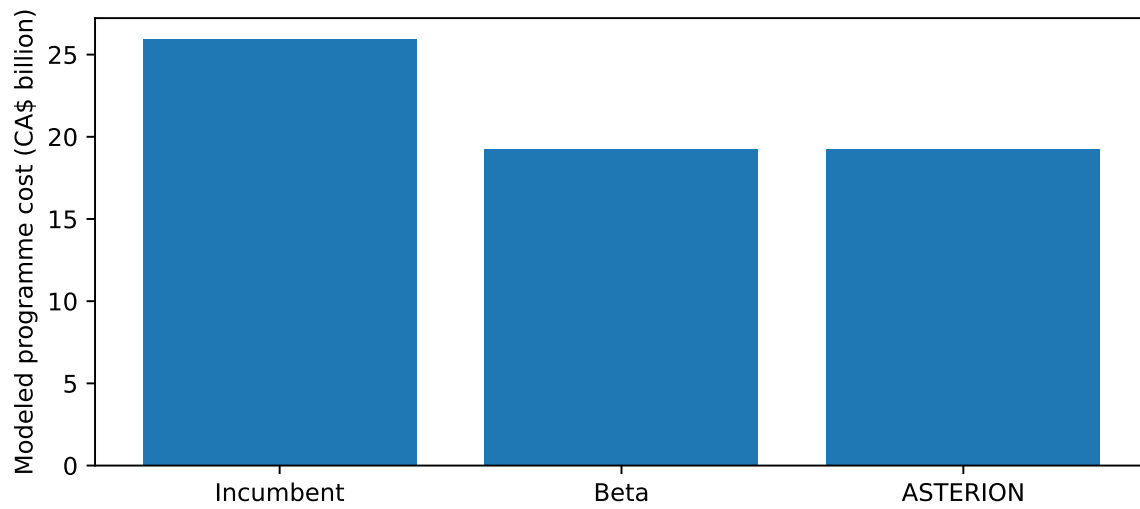


Figure 3. One-programme modeled costs. The large incumbent contrast must not obscure the much smaller candidate–Beta difference.

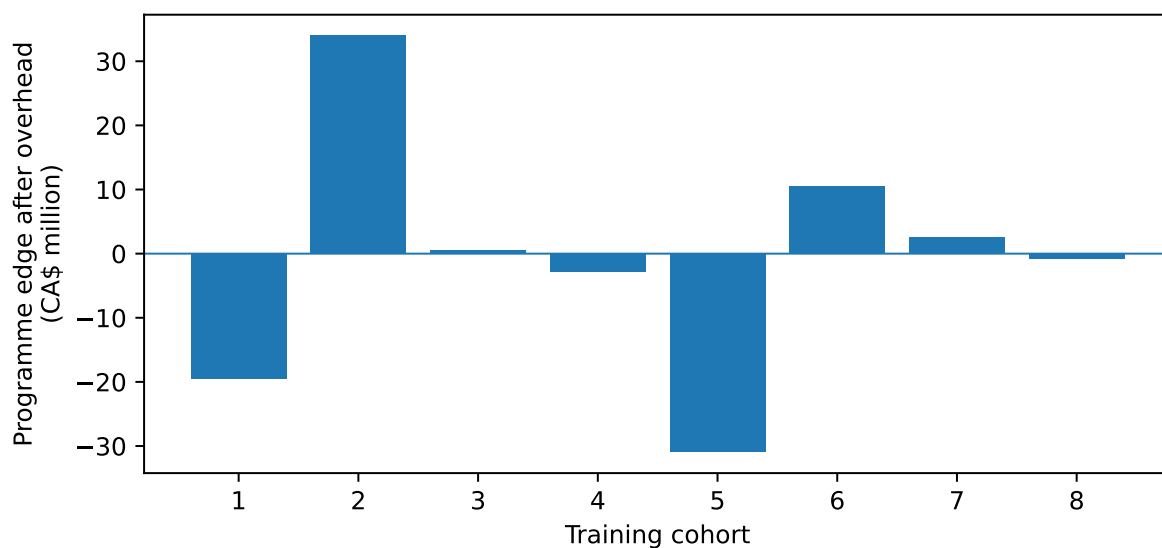


Figure 4. Eight cohort-level effects after overhead. Four are positive; the bootstrap interval crosses zero. These bars are not 2,048 independent training trials.

7.1 What the interval does and does not mean

The retained analysis resamples eight cohort mean effects using 10,000 bootstrap draws. Its conditional 95% interval is [CA\$ -13.325, CA\$ 12.085] million. The interval reflects the empirical distribution of those eight training cohorts in the authored world. It is not a distribution-free confidence guarantee, a bound on every future process, or an independent customer proof. Model selection history and common designer custody are not removed by a bootstrap.

7.2 Adversarial evidence changes the admission decision

All eight selected equations pass the nominal local threshold. All 6,144 accounting reconstructions agree, and all 672 injected ledger faults are detected. However, replacing the material response by

$$\varepsilon_{\text{shift}}(u, d) = \varepsilon_{\star}(u, d) - 0.081\{d > 1.4\} \quad (7.2)$$

creates 599 initial shortfalls. They are not excused by post hoc paid area. If the mission's hard requirement is sufficient capacity at first deployment, the shifted regime fails. The current decision is therefore **nominal local model pass, competitive economic no-go, adverse transfer failure**. [2]

8 Alpha as a simultaneous decision certificate

NEW MATHEMATICAL CONDITION / NOT SATISFIED BY ASSERTION

A useful operational definition compares full mission utility with fixed feasible alternatives. Let $Z_{i,b}$ be the paired advantage of frozen candidate c over comparator b on independent challenge unit i , before declared incremental deductions K . Fix m comparators and let each $Z_{i,b} \in [a_b, b_b]$. The notation b_b is the upper bound, not a baseline label.

Proposition 8.1 (Finite-comparator lower certificate). If the challenger, comparators, ranges and protocol are fixed before protected case access, the challenge units are independent and identically distributed, and K is fixed, then with probability at least $1 - \alpha$,

$$\min_b \mathbb{E} Z_{i,b} - K \geq \min_b \left\{ \bar{Z}_b - (b_b - a_b) \sqrt{\frac{\log(m/\alpha)}{2n}} \right\} - K. \quad (8.1)$$

This is a one-sided Hoeffding bound and union bound, with no independence required *between comparators* on the same unit. [13] It is intentionally conservative. The independent unit must match the claim: episodes, production batches, sites or independently fitted cohorts cannot be interchanged. The present eight-cohort bootstrap is not this certificate.

A numerical lower bound alone is insufficient. Rights, evidence custody, physical constraints, security, recovery and accountable authorization are conjunctive gates, not weighted score components. The admission predicate is

$$\text{Admit}(c) = 1\{\text{LCB}_c > h\} \prod_{j=1}^J 1\{G_j(c) = \text{pass}\}. \quad (8.2)$$

Across successive releases, allocating α_g with $\sum_g \alpha_g \leq \alpha_{\text{total}}$ controls the union of covered false-admission events under the assumptions of each trial. It does not permit adaptive reuse of leaked holdouts. The owner must also refresh Beta when the market or internal method improves: a historical spread cannot authorize a present monopoly premium.

9 Strategic verification: the inspection contract

PROPOSED GAME-THEORETIC DESIGN / NO REAL CONTRACT EXECUTED

An institution with valuable claims attracts strategic behavior. A claimant may save effort by hiding a failure; a verifier may save effort by rubber-stamping; a sponsor may prefer a flattering comparator. Independent evidence custody and precommitted analysis reduce the opportunities for such behavior, but incentives still matter. The following model states exactly what an inspection instrument can establish.

9.1 Imperfect detection and collectible consequences

A risk-neutral claimant chooses honest reporting H or misrepresentation M . Relative to H , misrepresentation yields net private gain $g > 0$ before enforcement. The owner commits to audit probability q . Conditional detection rates are d for M and false-positive rate f for H , with $d > f$. An adjudicated, collectible amount S is enforced with probability e . Then

$$U(M) - U(H) = g - qe(d - f)S. \quad (9.1)$$

Proposition 9.1 (Inspection deterrence). Honest reporting is weakly optimal if $qe(d - f)S \geq g$. A feasible audit probability exists only if $e(d - f)S \geq g$. Strict inequality gives strict preference in this two-action model.

With purely illustrative $g = \text{CA\$}0.8$ million, $S = \text{CA\$}4$ million, $d = 0.90$, $f = 0.01$ and $e = 0.95$, the minimum is $q \approx 0.23655$. This is scenario arithmetic, not an estimated fraud probability or legal recommendation. Unlimited sanctions, perfect adjudication and collectible bonds must not be assumed when they do not exist.

False positives also impose a cost on honest participation. A sponsor minimizing total cost would include qfL_H for wrongful adverse outcomes, the cost of appeals, and qC_A for inspection itself. If $q_{\min} > 1$, the remedy is to reduce the exploitable gain, improve detection, narrow authority or reject the contract—not to claim that cryptographic signing fixes the incentive failure.

9.2 The verifier is also strategic

A verifier must not be paid merely because a candidate passes. Effort can instead be evaluated against separately adjudicated reference checks and actual custody obligations. Such checks must be disclosed as auditor controls, never mixed into customer-performance evidence. A corrupt coalition with gain g_C requires a separate residual detection and enforceability analysis; the single-claimant inequality does not make a system coalition-proof.

Future rents can support cooperation only while they are credible. If a deviation yields g now and causes loss of discounted continuation value δV with probability p , deterrence requires $g \leq p\delta V$. Reputation without costly consequences, persistent identity or independent detection may supply little of that value.

10 Calibrated beliefs, correlated evidence and coordination

10.1 A proper score rewards truthful beliefs, not certainty

Let $Y \in \{0, 1\}$ be a future, independently observed event, such as a specified batch meeting a predeclared threshold. An evaluator with belief p reports $r \in [0, 1]$ and receives quadratic score $S(r, Y) = s_0 - k(r - Y)^2$, $k > 0$. Then

$$\mathbb{E}_p S(p, Y) - \mathbb{E}_p S(r, Y) = k(r - p)^2 \geq 0. \quad (10.1)$$

This strictly proper score rewards reporting one's probability under risk neutrality and exogenous outcomes.[14] It does not make the belief correct, establish causal identification or remain incentive-compatible when the reporter can manipulate the outcome, the sample or the measurement apparatus. A proper score cannot replace source control.

10.2 A committee cannot average away shared failure

Suppose all verifiers fail together with probability ρ ; otherwise n odd verifiers have independent error probability p . The majority-error probability is

$$P_{\text{err}}(n) = \rho + (1 - \rho) \sum_{k=(n+1)/2}^n \binom{n}{k} p^k (1 - p)^{n-k} \geq \rho. \quad (10.2)$$

Figure 5 evaluates this declared mixture with $p = 0.1$ and $\rho = 0.02$. More correlated copies are not more independent evidence. Different prompts do not establish different custody, instruments, incentives or failure modes. The parameter ρ must be investigated; it is not estimated by this toy calculation.

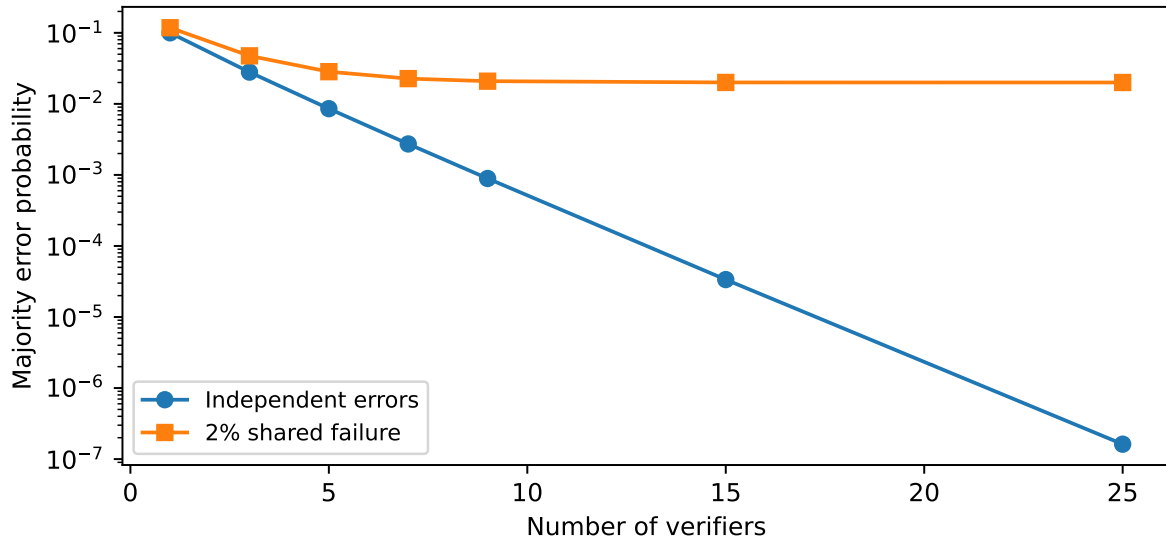


Figure 5. A declared 2% common-mode failure prevents committee error from vanishing. This is a model calculation, not an empirical verifier reliability estimate.

10.3 Correlated scheduling without confusing coordination and proof

A coordinator can improve scarce instrument allocation without controlling the scientific verdict. For finite actions and distribution $\mu(a)$, correlated-equilibrium obedience requires

$$\sum_{a_{-i}} \mu(a_i, a_{-i}) [u_i(a_i, a_{-i}) - u_i(a'_i, a_{-i})] \geq 0 \quad \forall i, a_i, a'_i. \quad (10.3)$$

This is the standard finite correlated-equilibrium condition.[15] In a two-laboratory scheduling example, each lab receives utility 2 when they use different time slots and 0 when they collide. Recommending (A, B) and (B, A) with equal probability gives utility 2 each and satisfies every obedience inequality. Independent uniform choices give expected utility 1 each.

The publication computes the four inequalities explicitly. This is a scheduling benefit, not independent validation or a declaration that the same mediator should control the candidate and its final verifier. Scientific custody can remain separated while calendars coordinate.

11 Research portfolios as potential games

FINITE MATHEMATICAL EXHIBIT / NOT AN EMPIRICAL LAW OF INSTITUTIONS

11.1 Aligning a laboratory with the portfolio

Let $s = (s_1, \dots, s_N)$ assign each laboratory to a mission or idle option. A disclosed portfolio potential $\Phi(s)$ values useful coverage and subtracts effort and congestion. The marginal-contribution utility

$$u_i(s) = \Phi(s) - \Phi(s_i^0, s_{-i}) \quad (11.1)$$

satisfies the exact potential identity

$$u_i(s'_i, s_{-i}) - u_i(s_i, s_{-i}) = \Phi(s'_i, s_{-i}) - \Phi(s_i, s_{-i}). \quad (11.2)$$

This aligns unilateral deviations with the declared potential.[16] It does not establish that Φ measures true scientific value, that utilities are budget-balanced payments, or that any local equilibrium is globally optimal. Feasible actions must still respect rights, experiment safety and resource constraints.

11.2 Asynchronous logit and the exact Gibbs identity

On a finite product action space, select one laboratory with fixed positive probability and redraw its action with

$$P_i(a_i | s_{-i}) = \frac{\exp(u_i(a_i, s_{-i})/\tau)}{\sum_{b_i} \exp(u_i(b_i, s_{-i})/\tau)}, \quad \tau > 0. \quad (11.3)$$

Proposition 11.1 (Finite logit stationary measure). For an exact potential game on a connected finite product space with these asynchronous updates, the chain is reversible and has unique stationary distribution

$$\pi_\tau(s) = Z_\tau^{-1} \exp(\Phi(s)/\tau). \quad (11.4)$$

For any probability mass function p , with Shannon entropy $H(p) = -\sum_s p(s) \log p(s)$,

$$[\mathbb{E}_{\pi_\tau} \Phi + \tau H(\pi_\tau)] - [\mathbb{E}_p \Phi + \tau H(p)] = \tau D_{\text{KL}}(p \| \pi_\tau) \geq 0. \quad (11.5)$$

The connection between stochastic strategic revision and statistical mechanics is established in the literature.[17, 16] Appendix E proves the specialized finite statement by detailed balance and direct expansion. If hard constraints disconnect the unilateral-move graph, uniqueness is only component-wise; unsupported global exploration must not be claimed.

Here τ has payoff units and H is dimensionless. It is not radiator temperature, heat or physical entropy. Equation (11.5) is an exact mathematical identity, not a physical law that makes an institution become truthful. In particular, concentration near a high potential can intensify exploitation of a misspecified scientific objective.

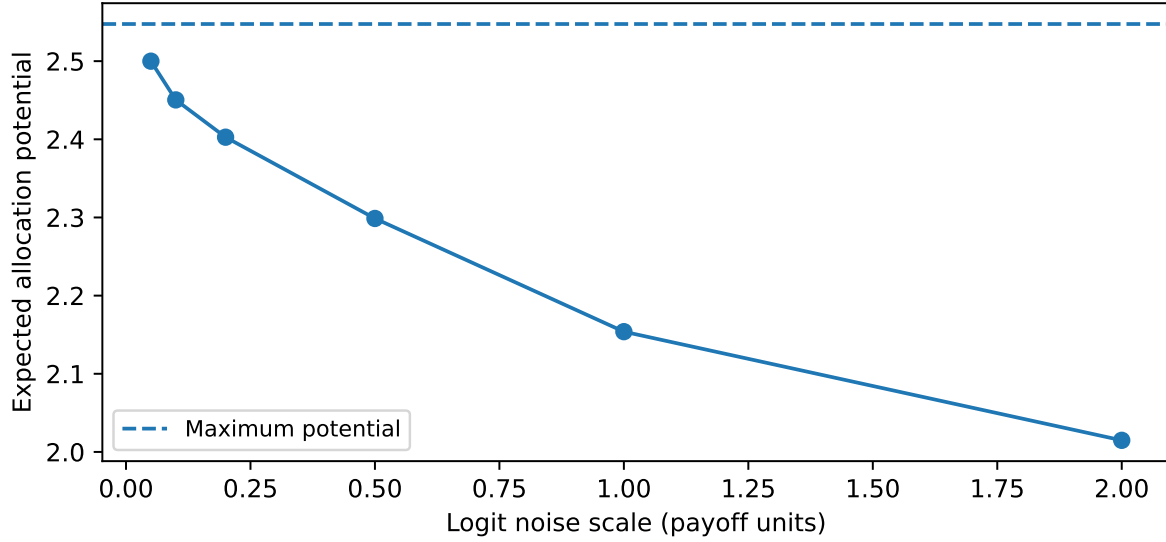


Figure 6. Four laboratories, two research missions and idle: 81 states. The publication checks 972 unilateral potential identities and six stationary measures. No laboratory behavior is empirically estimated.

12 Physical safety and institutional succession

12.1 A robust thermal barrier

For a proposed lumped thermal system $C\dot{T} = P_{\text{load}} + P_{\text{abs}} - Q_{\text{reject}}(T, u, \theta)$, let the safe set be $T \leq T_{\text{max}}$ and $h(T) = T_{\text{max}} - T$. Under a stated uncertainty set Θ , a sufficient differential condition is

$$\inf_{\theta \in \Theta} Q_{\text{reject}}(T, u, \theta) \geq P_{\text{load}} + P_{\text{abs}} - \kappa C(T_{\text{max}} - T), \quad \kappa > 0. \quad (12.1)$$

It implies $\dot{h} \geq -\kappa h$ and forward invariance when the dynamics are well posed, the condition is enforced continuously and feasible control exists. This is a thermal specialization of the control-barrier perspective.[18] Sampling, actuator delay, coolant constraints, uncertain absorption and unmodeled degradation can invalidate those premises. A controller must derate or enter a separately qualified safe mode when the constraint is infeasible; it must not optimize past an empty feasible set.

12.2 Authority is indexed by the exact release

Let a proof-bearing tuple be $\chi = (m, r, b, \mathcal{D}, \pi, v, \mathcal{E}, t_{\text{exp}})$: mission, release, baseline, evidence commitment, protocol, verifier, environment and expiry. Permission for action a requires every component to agree with the current operating state, together with the customer's policy and resource conditions.

$$\text{Allow}(a, t) = \mathbf{1}\{\text{Match}(\chi, t)\} \mathbf{1}\{t < t_{\text{exp}}\} \mathbf{1}\{a \in \mathcal{A}_{\text{policy}}(t)\} \mathbf{1}\{\text{RecoveryReady}(t)\}. \quad (12.2)$$

A descendant with $r' \neq r$ fails the match until new evidence and admission bind r' . It may inherit data, code, hypotheses and negative examples where rights permit; it does not inherit proof or authority. This is a formal restatement of the source doctrine, not a new claim that the ASTERION extension executed a live policy gateway.[1]

The owner sees capability, proof and authority as distinct state fields. A local model pass cannot be displayed as a production permit. A visually impressive interface must not show an Alpha claim as current while the comparative evidence is negative or expired. The purpose of institutional continuity is to preserve the ability to reconsider, not to preserve an obsolete answer.

13 From a new process advantage to productive capital

SEPARATE CONDITIONAL UNDERWRITING / NOT THE EXECUTED TRIAL'S RETURN

Valcourt's next high-value mission is not to relabel a near-tie as an exclusive breakthrough. It is to establish a genuinely new materials or process option—less degradation, lower unit cost, better yield, more robust operation—and compare it with refreshed Beta on equal feasible terms. The following ledger asks what such a future improvement would have to earn.

13.1 The conditional 2.5% case

Let $B = \text{CA\$ } 19.233$ billion denote the retained modeled Beta programme cost. A *new assumed* fractional advantage s yields gross room Bs . Deduct the following first-programme allowances and reserves, whose total is CA\$ 283 million.

Deduction	CA\$ million
Plant changes	110
External tests and qualification	20
Model, operator and maintenance	12
Transition work	18
Warranty-risk reserve	50
Imitation and drift reserve	70
Initial research allowance	3
Total	283

$$\Pi(s) = Bs - 283 \times 10^6, \quad s_{\text{break-even}} = \frac{283 \times 10^6}{B} = 0.014714 \dots \quad (13.1)$$

At $s = 0.025$, the conditional contribution is CA\$ 197.827 million. This is not a confidence bound, independently proven Alpha, a vendor quote or a tax calculation. No 2.5% improvement was discovered in the experiment. The expenditure schedule and uncertainty of benefit need separate underwriting before any commitment.

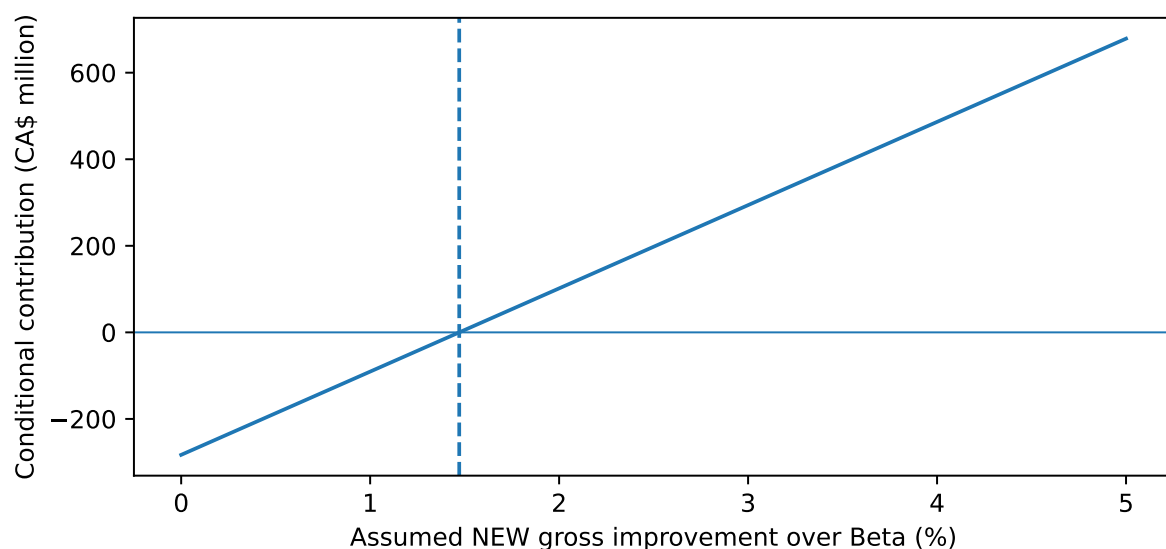


Figure 7. Separate hypothetical first-programme contribution. The crossing at 1.4714% is an assumption threshold, not an estimated probability of success.

13.2 Only accepted cash can finance the next capability

An invented conversion factor of 60% turns positive contribution into CA\$ 118.696 million of conditional available cash. The scenario assigns one half to replication/manufacturing, one quarter to science and proof, and one quarter to liquidity: CA\$ 59.348, CA\$ 29.674 and CA\$ 29.674 million. Negative contribution creates no positive reinvestment allocation.

These allocations do not purchase a factory whose cost exceeds the available envelope. They create a choice: accumulate accepted proceeds, secure separately approved financing, narrow the project or wait. Expected value and discounted valuation are not funds available for expenditure.

13.3 A resource-bounded capacity equation

Let B_t be cash, K_t commissioned productive capacity, z_t commitments to new capacity and ℓ a commissioning lag. In consistent time units,

$$\begin{aligned} B_{t+1} &= B_t + R_t^{\text{accepted}} + F_t - I_t - O_t - D_t, \\ K_{t+1} &= (1 - \delta)K_t + z_{t-\ell}, \\ z_t &\leq \min\{I_t/\kappa, M_t/\mu, E_t/e_K, \bar{F}_t, \bar{D}_t\}. \end{aligned} \tag{13.2}$$

Here F_t is explicitly authorized financing, O_t operating/research outflow, D_t distributions, κ capital per capacity unit, μ material intensity, e_K construction-energy intensity, and the barred quantities are fabrication and demand/commissioning limits. They are constraints in physical units, not monetary confidence scores. Instruments, skilled operators and raw materials cannot be synthesized by changing a portfolio valuation.

Proposition 13.1 (Positive growth mode with commissioning delay). In an ideal unsaturated continuous approximation, let $\dot{K}(t) = aK(t - \ell) - \delta K(t)$ with $a, \delta > 0$, $\ell \geq 0$. The real exponential growth mode satisfies

$$\lambda + \delta = ae^{-\lambda\ell}. \tag{13.3}$$

A positive real solution exists uniquely if and only if $a > \delta$. When it is positive, increasing ℓ decreases it.

For hypothetical $a = 0.08$ and $\delta = 0.03$ per year, the positive mode declines from 5% with zero lag to about 2.18% with twenty years of lag. Those are unsaturated mode calculations, not forecasts. Resource ceilings, opportunity erosion and finite demand prevent an unrestricted exponential extrapolation.

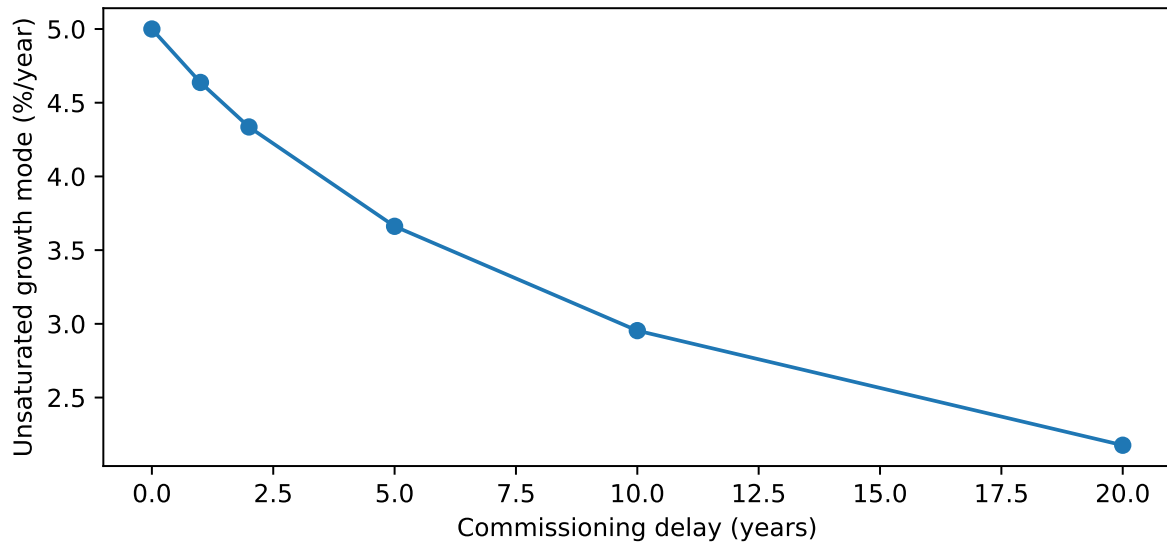


Figure 8. Commissioning lag reduces the positive real growth mode in the declared linear delay model. No stellar arrival date is inferred.

14 The stellar horizon: explicit power, heat, matter and latency

PROSPECTIVE RESOURCE ENVELOPE / NOT A CONSTRUCTION PLAN

The long-horizon use case is an estate capable of manufacturing its next scientific capability: more discriminating instruments, more reliable materials, more productive fabrication, more energy and more inspectable decision systems. Its distant form could include distributed collectors, computation, manufacturing, observatories and remote assembly. The aspiration is physically consequential; it is not a claimed outcome of owning the present software.

14.1 A consistent power account

Using IAU nominal solar luminosity $L_{\odot}^N = 3.828 \times 10^{26}$ W and nominal irradiance at 1 AU of 1361 W/m², the ideal projected collector area and stellar fraction for intercepted radiant power P are

$$A_{\text{collect}} = \frac{P}{1361 \text{ W/m}^2}, \quad f_{\star} = \frac{P}{L_{\odot}^N}. \quad (14.1)$$

These IAU values are exact nominal conversion standards, not instantaneous measurements.[19] If useful electrical power is required, a separately specified conversion efficiency and parasitic budget are needed. If that electricity is consumed locally, it eventually contributes to the estate's heat burden; it cannot be subtracted as though exported.

Illustrative $P = 10^{26}$ W envelope	Computed magnitude
Nominal stellar fraction	26.1%
Projected collection at 1 AU	$7.35 \times 10^{22} \text{ m}^2$
Effective emission, 450 K, $\varepsilon = 0.85$, 3 K sink	$5.06 \times 10^{22} \text{ m}^2$
Film-only mass at assumed 0.1 kg/m ²	$1.24 \times 10^{22} \text{ kg}$

The mass calculation adds collector and radiator areas and applies an invented common areal density. It excludes support, shielding, coolant, transmission, transport, control and repair systems. At substantial interception fraction, mutual shadowing, incoming irradiation and geometric consistency cannot be neglected. The scalar area estimate is a screening calculation, not evidence that a star-scale system is manufacturable.



Figure 9. Conceptual industrial estate, generated artwork. The image specifies no feasible orbit, deployment geometry, temperature, material or construction method.

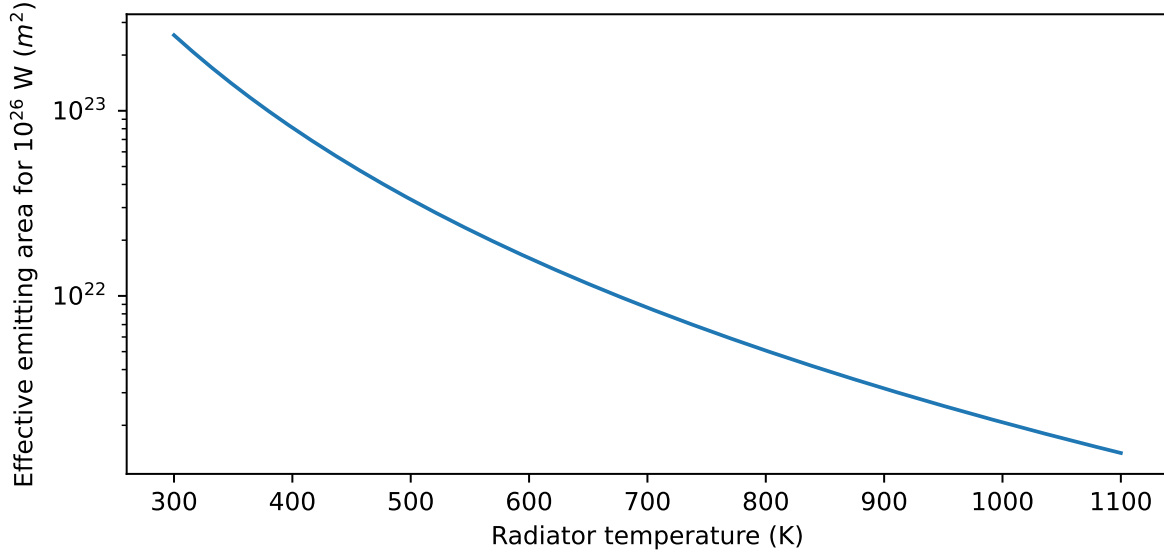


Figure 10. Ideal effective emitting area at fixed handled power, emissivity and sink. Higher temperature reduces area but requires separately qualified materials, conversion systems and thermal transport.

14.2 Area reduction is not free physical improvement

In the ideal cold-sink approximation, $A \propto T^{-4}$ and $\partial \log A / \partial \log T = -4$. Raising the radiator from 450 to 600 K reduces this modeled area to about 31.6% of its original value. But a design utility must also include material degradation, allowable temperatures, work-conversion losses and heat transport. Maximizing temperature without those constraints is another form of objective exploitation.

For a radially illuminated thin element with areal density Σ and radiation-pressure factor $\xi \in [0, 2]$, a simple force comparison gives

$$a_{\text{rad}} = \frac{\xi L}{4\pi r^2 c \Sigma}, \quad \beta_{\text{rad}} = \frac{a_{\text{rad}}}{GM/r^2} = \frac{\xi L}{4\pi c GM \Sigma}. \quad (14.2)$$

Making a film arbitrarily light changes its orbital dynamics; it is not merely a reduction in a mass ledger. This relation assumes radial illumination and omits orientation, drag, shadowing and active control. A credible assembly mission needs its own dynamical and reliability model.

14.3 Finite signal speed changes the governance topology

If a safety response requires a round trip across distance D and local computation/actuation time t_a , a necessary timing condition is

$$\frac{2D}{c} + t_a \leq \tau_{\text{safe}}. \quad (14.3)$$

A delay-sensitive local interlock cannot depend on a distant committee completing an impossible round trip. Distributed authority leases, local hard limits and asynchronously reconciled evidence become engineering necessities. Larger physical scale therefore motivates a federation of bounded institutions, not one omnipresent controller.

15 The optimal owner programme and the route to admission

ACTIONABLE RESEARCH DESIGN / FUTURE GATES NOT YET PASSED

The next defensible mission begins with one observable bottleneck, not a promise to discover everything. For ASTERION it is a real thermal-surface or process intervention whose useful lifetime and cost can be measured. The owner requests an evidence package with calibrated instruments, traceable sample preparation, representative exposure and a separately governed comparison. The investigator must be allowed to return failure.

Gate	Decision evidence	What becomes permissible
Constitute	Mission, observables, property rights, comparator and explicit harm limits	Bounded evidence acquisition; no deployment
Form	Reproducible candidate programs, failed hypotheses, instrument uncertainty	Research and controlled experiment proposals
Freeze	Exact complete release, instrument/adapter versions, protocol and budgets	Protected comparative examination
Prove	Independent custody, fresh outcomes, current Beta, all hard gates and full costs	A scoped result or a failure receipt; no automatic authority
Admit	Named policy owner, target security, controls, incident process and recovery drill	Only the explicitly granted, expiring action envelope
Renew	Drift, changed economics, supplier changes, failure analysis and successor evidence	Fresh admission or contraction/retirement

A useful future Specialist ASI would be defined by the precise material family, observation contracts, process, domain, decision utility and validity window in which it earns superiority. It need not solve every science to be economically consequential. A high-value, repeatable narrow mission can finance more capable instruments and more ambitious successor missions, while each transfer remains a new proof obligation.[1]

The owner interface should make six questions answerable without reading a training log: What do we understand? What could falsify it? What decision changes? What does the best alternative cost? What authority exists now? What is the next accountable decision? The supplied bilingual Mission Command lab exposes the retained model, economic no-go, conditional frontier and physical horizon. It is a local research interface, not a signed production cockpit.

16 Limitations and the productive thesis

The experiment is deliberately synthetic and representable within supported operators. It demonstrates a workflow rather than a new material. The 64 observations labeled “stress” in native format are nominal samples; the separate degradation shift supplies the genuine adverse-regime test and fails it. The residual rule is heuristic, the domain screen finite, the comparator set nonexhaustive, and the confidence interval based on only eight cohorts. Same-host replay does not create independent custody.

The new physics propositions require their stated assumptions; they are not model-checked production firmware. The game-theoretic mechanisms assume specific action sets, beliefs, observability and enforcement. Coalition resistance, false positives, uncollectible guarantees, misspecified portfolio value and common-mode failure remain real design problems. The capital model is resource-constrained but still authored, and no horizon calculation establishes a construction schedule.

What the institution is for

The objective is not to preserve an impressive model. It is to preserve a disciplined capacity to turn the next important uncertainty into a test, the next useful result into a better decision, and the next accepted surplus into more capable instruments and productive assets.

ASTERION’s present achievement is concrete: neural-guided symbolic programs are trained, revised, used in decisions, compared and retained. Its present commercial verdict is also concrete: the observed proprietary Alpha claim does not pass. The future opportunity is to produce a genuine new material/process advantage and earn its admission. The long horizon is an expanding scientific industrial estate whose growth remains accountable to physics, competitive evidence and available resources.

Intelligence becomes a durable asset when the institution can keep discovering what is worth building—and retain the evidence needed to build the next generation better.

A Notation and units

Symbol	Meaning	Units
u, d	Recipe and exposure in the authored material model	1
$\varepsilon, \hat{\varepsilon}, \underline{\varepsilon}$	Actual, predicted and lower-design emissivity	1
T, T_s, T_0	Surface, radiative sink and reference-environment temperatures	K
Q, P	Heat-rejection demand; separately identified radiant or electrical power	W
A, a	Effective emitting area; normalized area	m ² ; 1
$p(u)$	Authored area price	CAD/m ²
$C_i, \mathcal{V}$	Heat capacity of node i ; thermal availability	J/K; J
F_{ij}, J_i	Radiation view factor; surface radiosity	1; W/m ²
$Z_{i,b}, K, h$	Paired utility advantage, deductions and admission hurdle	mission units
q, d, f, e, S, g	Inspection probability, sensitivity, false-positive rate, enforcement, collectible consequence, temptation; d here is not material exposure	declared
Φ, τ, H	Portfolio potential, logit noise scale and Shannon entropy	payoff; payoff; nat
B_t, K_t, ℓ	Cash, commissioned capacity and commissioning lag	CAD; capacity; time

The same letter may occur in separate conventional domains only with an explicit local definition. Thermodynamic entropy, Shannon entropy and a monetary reserve are not interchangeable quantities.

B Proofs of the area and regret conditions

B.1 Conservative area

Proof of Proposition 5.1. The uniform error premise gives $\varepsilon - 2\delta \leq \underline{\varepsilon} \leq \varepsilon$. All terms are positive because $\varepsilon \geq e_{\min} > 2\delta$. Hence $\varepsilon/\underline{\varepsilon} \geq 1$, which gives feasibility. Moreover

$$\frac{A_R}{A_*} = \frac{\varepsilon}{\underline{\varepsilon}} \leq \frac{\varepsilon}{\varepsilon - 2\delta} \leq \frac{e_{\min}}{e_{\min} - 2\delta},$$

since $x/(x - 2\delta)$ is decreasing for $x > 2\delta$. No statistical estimation step appears in this proof; the uniform premise must be supplied separately. $\square$

B.2 A finite-design cost bound

Proof of Proposition 5.2. For each u ,

$$|\widehat{C}(u) - C(u)| = \frac{p(u)Q}{s_T} \frac{|\hat{\varepsilon}(u) - \varepsilon(u)|}{\hat{\varepsilon}(u)\varepsilon(u)} \leq M := \frac{p_{\max}Q\delta}{s_T e_0^2}.$$

Writing $u_* \in \arg \min C$, optimality of $\hat{u}$ gives

$$C(\hat{u}) \leq \widehat{C}(\hat{u}) + M \leq \widehat{C}(u_*) + M \leq C(u_*) + 2M.$$

An additive optimization tolerance η adds η to the regret bound. Different action menus or an erroneous feasible set do not satisfy this argument. $\square$

C Thermodynamic and barrier derivations

C.1 Dissipation of availability

Differentiate Equation (4.7), substitute conduction-only Equation (4.6), and pair the (i, j) and (j, i) terms:

$$\begin{aligned}\dot{V} &= \sum_i \left(1 - \frac{T_0}{T_i}\right) \sum_j K_{ij}(T_j - T_i) \\ &= \sum_{i < j} K_{ij}(T_j - T_i) T_0 \left(\frac{1}{T_j} - \frac{1}{T_i}\right) \\ &= -T_0 \sum_{i < j} K_{ij} \frac{(T_i - T_j)^2}{T_i T_j}.\end{aligned}$$

For general pairwise heat inflow $q_{ji} = -q_{ij}$ directed from hot to cold, the paired contribution is $T_0 q_{ji}(1/T_j - 1/T_i) \leq 0$. With external powers, the availability balance gains $\sum_i (1 - T_0/T_i)p_i$. Thus a positive supply term is priced as a physical input, not mistaken for spontaneous improvement.

C.2 Blackbody exergy algebra

In the specified blackbody reference construction, subtract the environmental energy flux and the environmental-temperature multiple of net entropy flux:

$$\begin{aligned}\dot{X} &= \sigma(T^4 - T_0^4) - T_0 \frac{4}{3} \sigma(T^3 - T_0^3) \\ &= \sigma T^4 \left[1 - \frac{4}{3}(T_0/T) + \frac{1}{3}(T_0/T)^4\right].\end{aligned}$$

No graybody entropy formula is obtained by simply multiplying this expression by emissivity. The boundary radiation field and optical modes must be specified before exergy is applied to an actual converter.

C.3 Thermal forward invariance

Equation (12.1) gives $\dot{h} \geq -\kappa h$ for every modeled uncertainty. Multiplying by $e^{\kappa t}$ yields $\frac{d}{dt}(e^{\kappa t} h(t)) \geq 0$, so $h(t) \geq h(0)e^{-\kappa t} \geq 0$ when $h(0) \geq 0$. The statement assumes a well-posed continuous trajectory, feasible enforcing control and no omitted disturbance outside Θ . It supplies no permission to ignore sample/hold error or actuator saturation.

D Statistical and strategic derivations

D.1 Simultaneous lower bounds

For fixed comparator b , Hoeffding gives

$$\Pr \left\{ \bar{Z}_b - \mathbb{E} Z_b > (b_b - a_b) \sqrt{\frac{\log(m/\alpha)}{2n}} \right\} \leq \alpha/m.$$

A union bound across m comparators yields simultaneous validity at least $1 - \alpha$. Taking the minimum across baselines and subtracting fixed K proves Proposition 8.1. If ranges or deductions were selected to favor the observed result, this proof would not apply.

D.2 Inspection and truthful reporting

Let honest reporting yield baseline payoff w before expected wrongful enforcement, and misrepresentation yield $w + g$ before expected valid enforcement. Then $U(H) = w - qefS$ and $U(M) = w + g - qedS$, giving Equation (9.1). The model assumes the stated gain and enforceability, not an arbitrary unlimited penalty.

For the quadratic score, $\mathbb{E}_p(r - Y)^2 = (r - p)^2 + p(1 - p)$. Subtracting the truthful report cancels $p(1 - p)$ and gives Equation (10.1). This elicits beliefs under the stated utility, not truth independent of measurement.

E Finite Gibbs derivation and its boundary

For states s, t differing only in player i 's action, the asynchronous selection probability of i and the conditional-logit denominator are identical in both directions. The potential identity therefore yields

$$\frac{P(s, t)}{P(t, s)} = \exp\left(\frac{\Phi(t) - \Phi(s)}{\tau}\right) = \frac{\pi_\tau(t)}{\pi_\tau(s)}.$$

Detailed balance follows. Positivity on the finite connected product space gives irreducibility and self-transitions give aperiodicity, establishing the unique stationary distribution. For the variational identity, expand

$$D_{\text{KL}}(p \| \pi_\tau) = \sum_s p(s) \log p(s) - \frac{\mathbb{E}_p \Phi}{\tau} + \log Z_\tau.$$

Since $\mathbb{E}_{\pi_\tau} \Phi + \tau H(\pi_\tau) = \tau \log Z_\tau$, Equation (11.5) follows. Vanishing τ concentrates stationary mass on global potential maxima in the finite connected case, but may make mixing very slow. This says nothing about the truth of the learned world or the independence of its evaluator.

The computed exhibit has four laboratories, actions 0, 1, 2 and

$$\Phi(s) = 3(1 - e^{-n_1(s)}) + 2.6(1 - e^{-n_2(s)}) - \sum_i c_i \mathbf{1}\{s_i \neq 0\}, \quad c = (.48, .57, .64, .73).$$

It checks all 972 unilateral action comparisons over 81 states; stationary, reversibility and variational residuals are evaluated for six positive noise scales. No equilibrium utilities are represented as monetary returns.

F Delayed productive reproduction

Substitute $K(t) = K_0 e^{\lambda t}$ into the ideal delay equation to obtain Equation (13.3). Define $f(\lambda) = \lambda + \delta - ae^{-\lambda\ell}$. Its derivative $f'(\lambda) = 1 + a\ell e^{-\lambda\ell} > 0$ makes the real root unique. Since $f(0) = \delta - a$ and $f(\lambda) \rightarrow +\infty$ as $\lambda \rightarrow +\infty$, the root is positive exactly when $a > \delta$. Implicit differentiation gives

$$\frac{d\lambda}{d\ell} = -\frac{a\lambda e^{-\lambda\ell}}{1 + a\ell e^{-\lambda\ell}} < 0 \quad (\lambda > 0).$$

The condition identifies a positive real mode in a linear unsaturated model. It is not a theorem of perpetual economic growth. Retirements, capacity constraints, cash delays, competing technologies and finite demand can dominate long before the formal mode is expressed at large scale.

G Complete cohort ledger

Values below are one-programme modeled advantages, in millions of Canadian dollars. Subtracting the additional three-million initial research allowance shifts each first-programme result by the same amount.

Cohort	Before overhead	After overhead	Shift failures	Selected Beta degree
1	−17.454	−19.454	69	2
2	36.011	34.011	68	2
3	2.488	0.488	82	2
4	−0.895	−2.895	89	2
5	−28.973	−30.973	65	4
6	12.574	10.574	75	2
7	4.642	2.642	74	3
8	1.156	−0.844	77	4
Mean	1.194	−0.806	599 total	

The eight independently generated training cohorts, not the 2,048 policy cases, are the resampling units for the retained interval. A positive model-level confidence interval under a fixed freeze would answer a different question from performance variability across new fitting cohorts. Both can matter to a customer; neither should replace the other without notice.

H Reproducibility, source map and checks

H.1 Read, inspect, reproduce

Open the package’s `START_HERE.html`. The PDF is the reading artifact; the LaTeX is the editable source. The original customer simulation is retained under `research/ASTERION_OMEGA_1_0`. Its own manifest, licence, negative result and interface remain intact. The publication’s new calculations are in `research/paper_analysis.py`.

Reproduction commands

```
python3 VERIFY_PACKAGE.py
cd research/ASTERION_OMEGA_1_0
python3 VERIFY_RELEASE.py
OPENBLAS_NUM_THREADS=1 OMP_NUM_THREADS=1 \
  python3 src/test_asterion.py
OPENBLAS_NUM_THREADS=1 OMP_NUM_THREADS=1 \
  python3 src/run_asterion.py --product product_snapshot \
  --out replay/new_run
```

Use a new output directory. Dependencies are not bundled; prepare the recorded Python 3.13 environment and approved packages in the retained requirements file. Windows shells use different environment-variable syntax. To recompute the paper’s analyses, run `python3 research/paper_analysis.py` from the publication root. Its charts require Matplotlib and its root calculation requires SciPy.

H.2 Fresh checks performed for this publication

The original 200-file capsule manifest verified successfully. All 24 customer-extension tests passed. The full eight-cohort experiment was rerun. Of its 95 tracked outputs, 94 match the retained bytes exactly; the

remaining product-binding file differs only in the displayed source-root name (`product_snapshot` rather than the earlier distribution directory), while its code hashes are unchanged. This path-label difference is disclosed rather than hidden in a byte-identity claim.

The publication-specific script passes 49 named checks, including recomputed cohort means and bootstrap interval, 972 potential deviations, six stationary/reversibility/variational tests, four correlated-obedience inequalities, inspection thresholds, common-mode floors, 2,000 thermal networks, 1,200 conditional area cases and six delay roots. These are finite numerical checks around proved or stated mathematical models, not a formal verification of the complete software stack.

The upstream report's 840 runtime and 106 research tests remain inherited records. Browser qualification in the original lab was partial; native local-file navigation was blocked by the authoring environment. This publication does not claim to close that operational gap. Hash agreement demonstrates integrity against a reference, not publisher authentication; independently trusted signatures would be a separate artifact.

H.3 Claim-to-artifact index

Claim or object	Inspection route inside the ZIP
Retained numerical result	research/ASTERION_OMEGA_1_0/ evidence/primary/SUMMARY.json
Every cohort and selected program	evidence/primary/COHORTS.json, cohort_00/ through cohort_07/
Individual design comparisons	evidence/primary/PAIRED_PLANS.csv
New mathematical calculations	research/paper_results/PAPER_ANALYSIS.json
Fresh experiment replay	assurance/EXPERIMENT_REPLAY.json
Rendered publication assurance	assurance/PUBLICATION_VALIDATION.json
Editable typesetting and figures	latex/, latex/figures/, artwork/

References

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- [2] MONTREAL.AI. *ASTERION Ω : Executed Simulation 1.0 and Complete Evidence Capsule*. User-supplied archive. SHA-256: 2516f5d59879c015e32d878aa70d788f17cd12036a40974946161ca033ba6a8d. Synthetic, same-author evidence. Sept. 5, 2026.
- [3] MONTREAL.AI. *NEURAL-SYMBOLIC SUCCESSOR Ω v10.2: Discovery Milestone Validation*. User-supplied release validation. Upstream tests are inherited exhibits, not new production qualification. Sept. 5, 2026.
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